

## Ushtrime ND8g

1.2.143 Unaza e telit të bakrit është e rëndë  $Q=10\text{ N}$ . Të llogaritet gjatësia e telit, nëse diametri i tij është  $d=1\text{ mm}$ ?

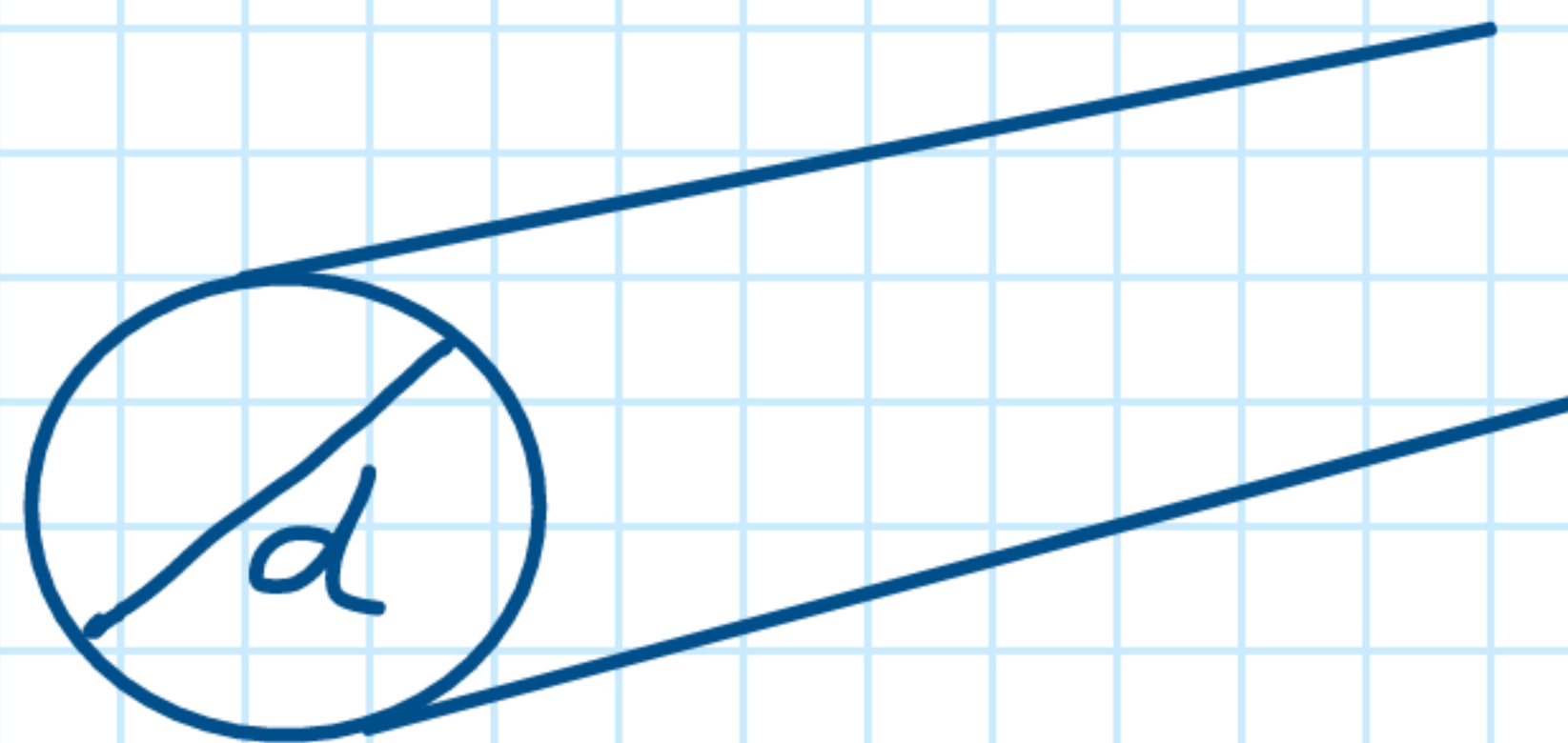
Zgjidhje:

Të dhënat:

$$Q=10\text{ N}$$

$$d=1\text{ mm}=1\cdot 10^{-3}\text{ m}$$

$$l=?$$



$$d=2r; r=\frac{d}{2}$$

$$S=\pi r^2=\pi \frac{d^2}{4}$$

$$F=\frac{Q}{V} \text{ or } F=g\cdot g$$

$\downarrow$   
 $V=\frac{Q}{F}$

$$V=S\cdot l \Rightarrow l=\frac{V}{S}=\frac{\frac{Q}{F}}{\pi \frac{d^2}{4}}=\frac{4Q}{\pi d^2 F}=\frac{4Q}{\pi d^2 \cdot g \cdot g}$$

$$l=\frac{4\cdot 10\text{ N}}{3,14\cdot 1\cdot 10^{-6}\text{ m}^2\cdot 9,81\frac{\text{m}}{\text{s}^2}\cdot 8900\frac{\text{kg}}{\text{m}^3}}=\frac{40\text{ N}}{0,27415\frac{\text{N}}{\text{m}^3}\text{ m}^2}=145,9\text{ m}$$

$$\rho_{\text{Cu}}=8900\frac{\text{kg}}{\text{m}^3}$$

Detyrat e Hidrostatikës

1.2.112, 113, 114, 115, 117, 118, 119, 120, 121, 123, 124

129, 134, 138, 140 dhe 143.